PES University - Dept. of CSE
Lab 1: Requirements Engineering & UML Use-Case Modelling
Problem Statement #43 - Feature Flag & Dynamic Config Manager

Use Case Flow Specification

Core use case detailed below: Evaluate Feature Flag.

Use Case Name	Evaluate Feature Flag
Actor	Client SDK
Description	The Client SDK requests the current state of a feature flag for a given user context, and the system returns a boolean result based on targeting rules, rollout percentage, and environment configuration.
Preconditions	1. The feature flag has been created and configured with at least a default state. 2. The client SDK holds a valid API key / connection to the flag service.
Main Success Scenario	1. The Client SDK sends an evaluation request containing the flag key and user context (User ID, attributes). 2. The system retrieves the flag's current configuration for the active environment. 3. The system includes the "Evaluate targeting rules" use case to check the user context against configured cohorts and rollout percentage. 4. The system determines the resulting boolean flag state. 5. The system returns the flag state to the Client SDK. 6. The Client SDK applies the returned state to gate the relevant feature in the application.
Alternate Flow	A1 - Backend unreachable: If the flag service cannot be reached at step 2, the Client SDK serves the last cached flag state instead of failing the request, and resumes live evaluation once connectivity is restored.
Postconditions	1. An accurate boolean flag state has been returned to the Client SDK within the 15 ms performance target (NFR-001). 2. No flag configuration data has been modified during evaluation (read-only use case).